

Null Propagation as the Causal-Limit Invariant: Why the Speed of Light Is a Metric Representative

Amos Jay Maley

Abstract

The speed of light in vacuum is widely described as a fundamental constant. This paper argues that the speed of light is not fundamental in the relevant structural sense: it is not primitive causal structure, but a metric-dependent representative of a deeper invariant. The causal-limit invariant traditionally represented by the speed of light is *null propagation*: limiting propagation along lightlike structure at the boundary between timelike causal reachability and spacelike separation. In standard classical relativistic field theory, and in the geometric-optics regime, vacuum light propagation locally realizes this boundary role; the numerical constant c is the metric representative assigned to that realization only after a metric convention has been fixed. The claim is not the elementary observation that dimensional quantities depend on units. It is the stronger structural claim that speed presupposes metricity, whereas the light-speed invariant is more accurately located at the level of null propagation, null structure, and causal ordering. We formalize a representation/invariance criterion for fundamentality, define metricity, null structure, and null propagation, and classify the light-speed invariant as a null-propagation invariant rather than a primitive numerical speed. A boundary-level reading then sharpens the result: the causal boundary is not itself a metric object, while metric speeds, rates, and numerical propagation magnitudes enter only after a metric representation has been admitted. Standard empirical tests—Lorentz invariance, dispersion bounds, and causal signaling constraints—therefore constrain null-propagation structure rather than a primitive numerical constant. The final distinction is between substantive modifications of relativity, which alter null propagation or the associated null structure, and merely representational changes often described as “varying c .”

1 Introduction

The speed of light occupies a central place in modern physics. In special relativity it is presented as invariant across inertial frames. In general relativity it shapes causal structure via null cones. In quantum field theory it appears in Lorentz-invariant formulations. In metrology it is assigned an exact numerical value by convention.

Given this prominence, it is common to regard the speed of light as a *fundamental* constant of nature. This paper argues that this framing misidentifies the invariant.

The thesis is precise:

The light-speed invariant of relativity is null propagation: the limiting boundary mode of causal propagation separating timelike causal reachability from spacelike separation. The numerical speed of light is the metric representative assigned to that boundary after metric conventions have been fixed.

This is not an empirical challenge to relativity. It is a structural clarification: what part of the theory is invariant, what part is representational, and what notion of “fundamental” is coherent under that distinction.

The contribution is not the discovery that null cones matter. That is standard relativistic geometry. The contribution is the explicit separation of three levels often conflated in discussions of c : null propagation as the causal-limit invariant, vacuum light propagation as a local physical realizer of that invariant in the standard classical/geometric-optics regime, and numerical c as the metric representative used after a representation has been fixed.

This claim does not deny that c can be adopted as a primitive parameter in a chosen formal presentation, for example as a conversion factor or invariant scale in a metric formulation. The claim is conditional on the representation-invariance criterion used below: the invariant physical content attributed to c is carried by null-cone/null-propagation structure rather than by a numerical speed magnitude.

The phrase “speed of causality” is therefore useful only as shorthand. Strictly, a speed is already a metric quantity. The deeper invariant is not a more fundamental velocity, but the null-propagation boundary that determines which relations lie within causal reachability, which lie outside it, and which lie exactly at the limiting boundary between the two.

2 Fundamentality as invariance under admissible re-description

We require an explicit criterion.

Definition 1 (Representation). A *representation* of a physical theory is a choice of descriptive apparatus sufficient to state and compute within the theory: coordinates, parameterizations, conventions, and unit assignments, together with any derived numerical fields introduced by those choices.

Definition 2 (Admissible re-description). An *admissible re-description* is a change of representation that preserves the theory’s physical content: empirical predictions, invariants, and the equivalence class of models. Typical examples include coordinate changes, reparameterizations, and changes of units.

Definition 3 (Fundamental structure). A structure or quantity is *fundamental* (for a given theory) if it is invariant under all admissible re-descriptions of that theory.

This criterion is standard in spirit: it treats physical structure as what survives representational changes.

Remark 1. This definition does not imply metaphysical necessity. It asserts only representational primacy: fundamental structure is not an artifact of description.

3 Metricity and the concept of speed

The central observation is not “ c has units.” It is that the *concept* of speed presupposes a metric.

Definition 4 (Metricity). A theory is *metric* (in the relevant sense) when it provides a mapping from physical relations to numerical magnitudes of length and duration, i.e. a rule that assigns numbers to spatial separations and temporal separations in a coherent way sufficient to define ratios such as speed.

A speed is a ratio

$$v = \frac{(\text{spatial distance})}{(\text{temporal duration})},$$

which is undefined without:

- (i) a notion of spatial distance,
- (ii) a notion of temporal duration,
- (iii) a stable mapping from physical events/processes to those numbers.

All three are representational commitments. Even in classical physics, numerical distances and durations depend on choice of units and conventions. In relativity, additional representational freedom exists (coordinates, foliations, parameterizations).

The question of fundamentality is therefore not whether light has an invariant role (it does), but whether the *numerical speed* assigned to that role is primitive.

4 Null structure and null propagation

Relativity can be presented in geometric terms as statements about causal structure.

Definition 5 (Causal classification). In a local Lorentzian representation, the timelike, null, or spacelike classification of a nonzero tangent vector v^a is determined by the sign of $g_{ab}v^av^b$ (up to the chosen sign convention). Timelike vectors lie inside the local light cone, spacelike vectors lie outside it, and null vectors lie on its boundary. For event pairs, the corresponding global language is given by future-directed timelike or causal curves wherever such global relations are defined. Thus the operational language of influence is used here as shorthand for the relevant local light-cone classification and its associated causal-reachability relations.

Definition 6 (Null structure). *Null structure* is the boundary relation that separates timelike and spacelike classes, together with its invariance across admissible descriptions. Equivalently, it is the family of null cones (or their abstract ordering properties) that determine causal reachability.

Definition 7 (Null propagation). *Null propagation* is propagation whose causal role lies exactly on the boundary between timelike causal reachability and spacelike separation. It is not defined by a numerical velocity. It is defined by its limiting boundary role within the causal classification.

Null structure is the invariant boundary; null propagation is the invariant causal role of propagation along that boundary. In standard classical relativistic field theory, and in the geometric-optics regime, light in vacuum locally realizes the null-propagation class. It does not define the class. The class is fixed by the causal-boundary structure of the theory; the numerical constant c is the metric expression assigned after representation has been chosen.

5 Main result

The central theorem is the following. It is not intended to derive null preservation from weaker assumptions. It classifies which part of the light-speed formalism belongs to the invariant equivalence class once physical-content-preserving re-description is fixed.

Theorem 1 (Classification of the light-speed invariant). *Let a relativistic theory admit at least one admissible metric representation in which vacuum light propagation is null in the standard classical/geometric-optics regime. Within the class of re-descriptions that preserve the theory's empirical and causal content:*

- (a) *the null structure is preserved;*

(b) *the null-propagation role is preserved;*

(c) *any numerical speed assigned to that role is representation-dependent.*

Hence the invariant usually expressed as light-speed universality is more accurately classified as a null-propagation, or causal-limit, invariant. The numerical speed of light is its metric representative rather than primitive causal structure.

Proof. (a) Null structure is a causal-boundary classification. Under the admissible re-descriptions considered here—coordinate transformations, reparameterizations, unit changes, and equivalent metric presentations—the physical model is not changed. Causal reachability and boundary relations therefore remain fixed even when their coordinates or numerical representatives change. In standard differential geometric formulations, nullness is preserved under diffeomorphisms and under positive conformal rescalings of the metric; in operational formulations, null structure is preserved because it is defined by invariant signal-boundary behavior.

(b) Null propagation is propagation occupying the null role in the preserved classification. Since admissible re-description preserves the boundary classification, it also preserves the role of limiting causal propagation along that boundary. Thus the invariant is not a number attached to propagation, but the fact that the propagation lies on the null boundary. Vacuum light propagation is one physical realization of this role inside standard classical relativistic physics in the geometric-optics regime.

(c) A numerical speed is a ratio of a numerical length assignment to a numerical time assignment. Under a change of units (e.g. rescaling the meter by a factor $\alpha > 0$ while leaving the second fixed, or vice versa), the numerical value of any speed rescales by α (or its inverse). Since unit changes are admissible re-descriptions, the numerical value of the speed is not invariant. Even beyond unit scaling, in curved spacetime coordinate choices can change coordinate speeds of null signals without altering nullness. Therefore the numerical speed is representation-dependent.

Therefore null propagation is the relevant causal-limit invariant under the invariance criterion, whereas numerical light-speed is a metric representative. $\square$

Remark 2. The theorem is a classification result, not a claim that null cones or conformal structure are new discoveries. It identifies a stronger logical point than the triviality “dimensional constants depend on units”: *speed is a derivative construct requiring metricity*, whereas the light-speed invariant can be stated at the level of null propagation without presupposing a primitive numerical speed.

6 The hierarchy: null propagation, light propagation, and c

The result can be summarized as a three-level hierarchy:

$$\text{null propagation} \longrightarrow \text{vacuum light propagation} \longrightarrow \text{numerical } c.$$

The first level is the invariant causal-limit role: limiting propagation along lightlike structure. The second level is the physical realization of that role by light in vacuum within standard classical relativistic physics, especially in the geometric-optics regime. The third level is the numerical metric encoding of that realization in a chosen representation.

Confusing these levels creates the false impression that a number is the source of causal order. It is not. The number c is indispensable in practical computation, unit systems, and metric formulations, but its indispensability as a representative does not make it primitive structure. What is primitive in the relevant representational sense is the preserved causal-limit role: propagation at the null boundary.

7 Boundary-level and metric-level structure

The same hierarchy can be expressed without treating the causal boundary itself as a metric object. A boundary-level causal classification separates timelike reachability, spacelike separation, and the null limit before any numerical speed is assigned, though not before the theory has fixed the conformal or light-cone structure that makes such classification meaningful. It does not by itself carry lengths, durations, rates, or scalar propagation magnitudes. Those quantities enter only after a metric representation has been admitted.

This is the role played by null propagation in the present paper. It is not a scalar propagation rate at the boundary. It is the boundary-status role of propagation lying exactly at the causal limit. Metric descriptors such as speed, coordinate velocity, and numerical c are therefore representation-level devices: indispensable once a metric representation is in use, but not the source of the causal classification.

The boundary-status reading also explains why null propagation should not be converted into a graded or scalar selector. A physically meaningful causal boundary is sharp: a propagation relation either occupies the null role in the relevant representation class or it does not. The appropriate invariant is therefore the boundary status itself, not a posteriorly selected coupling strength, coordinate speed, or numerical ranking.

No additional dynamics is introduced by this interpretation. It supplies a disciplined reading of the same relativistic structure: null propagation is the causal-limit invariant, and c is the metric representative used once the theory is expressed inside an admitted metric representation. This terminology is compatible with constructional accounts of AMetric boundary structure, licensed interior metric bookkeeping, and stable bivalent boundary traces, but the argument of the present paper does not depend on that broader formalism [7, 8].

8 How standard formulations fit this result

A standard objection is that special relativity *postulates* invariance of the speed of light, so the present analysis might appear to be only a rephrasing of the theory.

The correct relationship is:

“The speed of light is invariant across inertial frames” is an operational encoding of “the null-propagation role and the associated null structure are invariant across admissible inertial re-descriptions.”

In Minkowski space, Lorentz transformations preserve the light cone, i.e. they preserve the null classification. If one chooses a particular unit convention and coordinate chart, this invariance can be expressed as a numerical statement about c . But the numerical statement is a *representation* of a deeper invariant.

Thus the analysis preserves special relativity while locating the light-speed invariant at the level of null-cone/null-propagation structure rather than at the level of a primitive numerical speed.

9 Metrology and the SI definition of c

In the modern SI, the numerical value of c in m/s is fixed *by definition*. This invites a common confusion: that definitional exactness is evidence of physical fundamentality.

It is not.

Corollary 1 (Definitional fixation is not physical fundamentality). *Fixing the numerical value of c by unit convention does not elevate the speed of light to fundamental structure. It only fixes a bookkeeping parameter and relocates uncertainty into the practical realization of units.*

Proof. A unit convention is an admissible re-description. By the invariance criterion, adopting a convention cannot create new fundamental structure; it can only select a convenient representative of an equivalence class. The physical content remains in the invariants (e.g. null propagation and null structure), not in the chosen numerical value. $\square$

10 Empirical content: what experiments actually constrain

A second objection is that experiments “measure c .” The more precise statement is:

Experiments test properties of causal structure (Lorentz invariance, isotropy, absence of dispersion, absence of preferred frames), which can be represented as constraints on a numerical c *within a metric representation*.

10.1 Time-of-flight and operational definitions

Time-of-flight experiments compare:

- a clock process (defining duration),
- a ruler or length standard (defining distance),
- the propagation of a signal.

All three are physical standards interpreted within an operational metric representation. The empirical question is not whether c has a particular numerical value, but whether propagation occupies the null-propagation role defined by the theory and whether that boundary role is stable across inertial transformations.

10.2 Lorentz invariance tests

Michelson–Morley-type experiments and modern resonator tests constrain anisotropy and frame dependence. In geometric language: they constrain deformations of the light cone and violations of null-cone invariance. Such constraints attach directly to null propagation and null structure.

10.3 Dispersion bounds and effective velocities

Astrophysical dispersion constraints (e.g. arrival time differences across frequencies) test whether null propagation depends on energy/frequency, i.e. whether the null boundary is sharp and universal. Again, the constrained object is null-propagation structure, not a primitive speed.

In dispersive media or effective descriptions, phase velocity and group velocity need not coincide with the relativistic causal limit. They may be useful kinematic quantities without being the invariant at issue here. The relevant causal constraint is the limiting signal-front or null-cone structure that bounds causal propagation in the underlying relativistic theory. This is why the paper distinguishes null propagation from any particular measured, effective, coordinate, phase, or group speed.

Remark 3. Nothing in this section denies the utility of c as a parameter. The point is categorical: the parameter is a metric encoding of causal-boundary structure.

11 General relativity: coordinate speeds vs null invariance

In general relativity, the coordinate speed of light can vary depending on coordinate choices, even in vacuum regions. This is often presented as a conceptual complication. In the present analysis it is a direct confirmation of the representational status of “speed.”

What remains invariant is:

- local nullness (light follows null curves),
- causal ordering,
- the null cones determined by the spacetime geometry,
- the null-propagation role of limiting causal propagation.

Thus both SR and GR support the same structural conclusion: the invariant object is the null-propagation boundary, while speed is a coordinate-, unit-, and representation-dependent encoding.

12 Clarifying “varying- c ” discussions

Many discussions of “varying speed of light” theories conflate:

- (i) mere reparameterizations (unit rescalings, field redefinitions, coordinate choices) that change the numerical value of a speed, with
- (ii) substantive changes to causal structure (altered null cones, preferred frames, modified dispersion relations).

Only (ii) is physically substantive. A numerical “variation of c ” that does not alter null propagation or null structure is representational. A physically meaningful varying- c theory is not fundamentally a theory in which a number changes; it is a theory in which the null-propagation structure is altered.

Corollary 2 (Meaningful variation targets null propagation). *Any physically meaningful departure from relativistic light-speed universality must manifest as a change in null propagation or the associated null structure (light-cone deformation, dispersion, or preferred-frame effects), not merely as a change in a numerical constant under unit conventions.*

13 What this paper does *not* claim

To reduce misreading, we state explicitly what is not being asserted:

- The paper does *not* claim that relativity is false.
- The paper does *not* claim that light propagation is not universal in the relevant invariant sense.
- The paper does *not* claim that c lacks an invariant physical role; it claims that this role is more accurately located at the level of null-cone and null-propagation structure than at the level of a numerical speed magnitude.

- The paper does *not* claim that c cannot be taken as primitive in a chosen formal presentation. It claims only that, under the invariance criterion used here, numerical c is not the primitive bearer of causal structure.
- The paper does *not* claim that c is dispensable in practical computation or engineering.
- The paper does *not* propose a new dynamics or a new “varying- c ” model.

The claim is narrower and stronger: *numerical speed is not primitive causal structure; null propagation is the causal-limit invariant represented by c .*

14 Conclusion

The speed of light is not fundamental in the representation-invariance sense examined here. Numerical and coordinate speed representations depend on metric conventions, coordinate choices, and unit assignments. The invariant local role is instead carried by the null-cone/null-propagation structure those speed representations express: the limiting boundary mode of causal propagation separating timelike reachability from spacelike separation, preserved under admissible re-description.

This clarifies why definitional fixation of c in metrology is conceptually orthogonal to physical fundamentality, why coordinate speeds vary in general relativity without physical contradiction, and why the meaningful target of any departure from relativity is deformation of null propagation or null structure rather than a numerical constant.

A Conformal Invariance of Local Null Structure

This appendix makes explicit a standard result from general relativity that is invoked in the main theorem.

Lemma 1 (Conformal Invariance of Local Null Structure). *Let (M, g_{ab}) be a Lorentzian spacetime and let $\tilde{g}_{ab} = \Omega^2 g_{ab}$ with $\Omega : M \rightarrow \mathbb{R}^+$ smooth and strictly positive. A vector is null with respect to g_{ab} if and only if it is null with respect to $\tilde{g}_{ab}$, and a smooth curve is null when its tangent is null everywhere. Consequently, the local light-cone structure and causal classification into timelike, spacelike, and null relations relevant to null propagation are preserved under positive conformal rescaling. For causal curves within the same time orientation, the corresponding causal reachability relation is preserved under such rescalings.*

Proof. For any tangent vector v^a ,

$$\tilde{g}_{ab} v^a v^b = \Omega^2 g_{ab} v^a v^b.$$

Since $\Omega^2 > 0$, the sign of $g_{ab} v^a v^b$ is preserved. Thus null vectors remain null, timelike vectors remain timelike, and spacelike vectors remain spacelike. The local light cones and the null-propagation boundary are therefore invariant under positive conformal rescalings of the metric. $\square$

This result underwrites the claim in the main theorem that null structure is preserved under admissible re-descriptions in general relativity, while avoiding the stronger and unnecessary claim that every global causal feature follows from conformal rescaling alone without auxiliary assumptions.

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